

Advanced Econometrics

Instrumental Variables Estimation

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Outline

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A Quick Reminder: What is OLS?

Ordinary Least Squares (OLS) estimates the relationship between X and Y :

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

What OLS Does

It finds the line that **minimises the sum of squared residuals** — the vertical distances between the data points and your estimated line.

Why We Like OLS (The Gauss-Markov Theorem)

Under the “Classical Assumptions”, OLS is **BLUE**:

- **B**est — smallest possible variance among all linear estimators
- **L**inear — in the parameters
- **U**nbiased — on average, it gives the right answer
- **E**stimator

The Big Assumption OLS Needs

The most critical assumption for OLS to work is:

$$\text{Cov}(X_i, \varepsilon_i) = 0$$

In Plain English

The explanatory variable X must be **unrelated to the error term** ε . Everything that correlates with Y but is not in our model (and lives inside ε) must have **nothing to do with** X .

When Is This Plausible?

A classic “safe” example: Does rainfall (X) affect crop yields (Y)?

Rainfall is determined by weather patterns — it is very unlikely to be correlated with omitted farm-level characteristics. OLS is probably fine here.

The Fundamental Problem: Endogeneity

Definition: Endogeneity

A variable X is **endogenous** when it is **correlated with the error term**:

$$\text{Cov}(X_i, \varepsilon_i) \neq 0$$

When this happens, OLS is **biased and inconsistent** — it gives the wrong answer even with a huge sample.

Three common culprits:

- **Omitted Variable Bias (OVB):** You forgot an important variable that is related to both X and Y . wages/schooling/ability — the classic case where a hidden variable steals the credit
- **Measurement Error:** Your data for X is noisy or mismeasured. self-reported hours worked — explains the direction of bias (attenuation toward zero) in plain terms
- **Reverse Causality:** X causes Y , but Y also causes X (a loop!). chicken-and-egg problem

Endogenous vs Exogenous Variables

Endogenous Variable

“Inside the system” — its value is determined *within* the model and is correlated with the error term ε .

Problem: OLS cannot cleanly separate its effect on Y from background noise.

Examples:

- Years of schooling (correlated with unobserved ability)
- Price of a good (simultaneously determined by supply and demand)
- Cigarette consumption (correlated with unobserved health habits)

Exogenous Variable

“Outside the system” — its value is determined *outside* the model and is **not** correlated with the error term ε .

Safe to use in OLS. Variation in this variable is “clean”.

Examples:

- Rainfall (not caused by crop yields)
- Distance to a college (fixed by geography/history)
- Age (not determined by the outcome we study)

One-Line Summary

Exogenous $\Rightarrow$ OLS is fine. Endogenous $\Rightarrow$ you need IV, DiD, or another fix.

Concrete Example: Returns to Education

Suppose we want to estimate how much an extra year of schooling (S) raises wages (W):

$$\ln(W_i) = \beta_0 + \beta_1 S_i + \varepsilon_i$$

The Problem: Ability Bias

People with higher **innate ability** (A):

- tend to **stay in school longer** ($A \uparrow \Rightarrow S \uparrow$)
- also earn **higher wages** regardless ($A \uparrow \Rightarrow W \uparrow$)

But we **cannot measure ability**, so it sits inside ε .

Result: $\text{Cov}(S_i, \varepsilon_i) \neq 0$ — OLS will **overestimate** β_1 !

Intuition

OLS sees that high-schooling people earn more, but it cannot tell: is it *because of* the schooling, or *because* they were already smarter?

OLS vs IV: The Core Difference

OLS

Uses all variation in X to estimate β .

Good: Efficient, uses all data.

Bad: If any of that variation is “contaminated” by correlation with ε , the estimate is **wrong**.

IV (2SLS)

Uses only the “clean” part of variation in X — the part driven by an instrument Z .

Good: Consistent even with endogeneity.

Bad: Less precise (higher standard errors) if the instrument is weak.

Key Trade-off

IV “throws away” variation to fix bias. You pay in **precision**; you gain **consistency**. (Even though your standard errors are wide and your results might look “insignificant” in a small sample, you have successfully eliminated the bias.) **If the standard error is high, the resulting t-statistic will be low. This makes it much harder to reject the null hypothesis, meaning your results are more likely to be deemed statistically insignificant ($p > 0.05$).**

Estimator Performance: OLS vs. Instrumental Variables

The IV Precision Tax

IV “throws away” variation to fix bias. You pay in **precision**; you gain **consistency**.

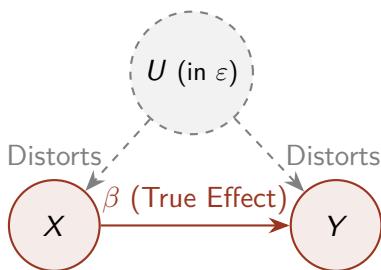
If the standard error is high, the resulting t-statistic will be low. This makes it much harder to reject the null hypothesis, meaning your results are more likely to be deemed statistically insignificant ($p > 0.05$).

Estimator	Bias (Small Sample)	Consistency (Large Sample)	Precision (Standard Error)
OLS	Biased	Inconsistent (converges to wrong answer)	Highly Precise (Low SEs)
IV	Biased	Consistent (converges to true answer)	Less Precise (High SEs)

Table: Comparison of OLS and IV asymptotic and finite sample properties.

Visualising the Endogeneity Problem

When a confounding variable (U) is omitted, it enters ε . This creates a **back-door path** from X to Y that OLS cannot separate from the true effect.

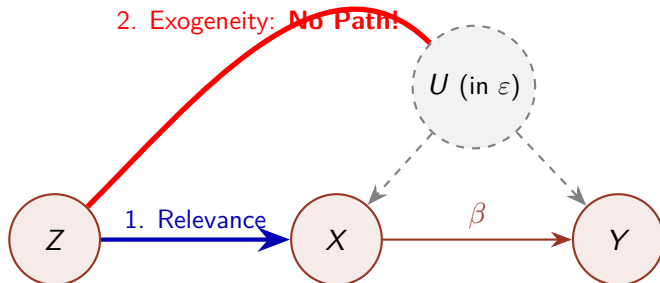


Why OLS Fails

OLS measures **both paths at once** — the true causal β *plus* the back-door path through U . It **cannot separate them**.

The Idea Behind an Instrument (Z)

An **Instrumental Variable** (Z) is a variable that acts like a “remote control” for X — it moves X but has no direct effect on Y , and no connection to the hidden U .



The Two Requirements for a Valid Instrument

- ① **Relevance:** Z must be correlated with X : $\text{Cov}(Z, X) \neq 0$.
- ② **Exogeneity:** Z must be uncorrelated with the error: $\text{Cov}(Z, \varepsilon) = 0$.

The second condition **cannot be tested** directly — it requires economic

A Classic Instrument: Card (1995)

Problem: We want the causal effect of schooling on wages, but ability is unobserved.

Instrument used: *Proximity to a college* (whether a 4-year college is nearby).

Relevance ✓

Growing up near a college **increases the probability of attending** (lower transport/moving costs).

$$\text{Cov}(\text{proximity}, S) \neq 0$$

Exogeneity (argued) ✓

College location is largely determined by **historical decisions**, not by individual ability.

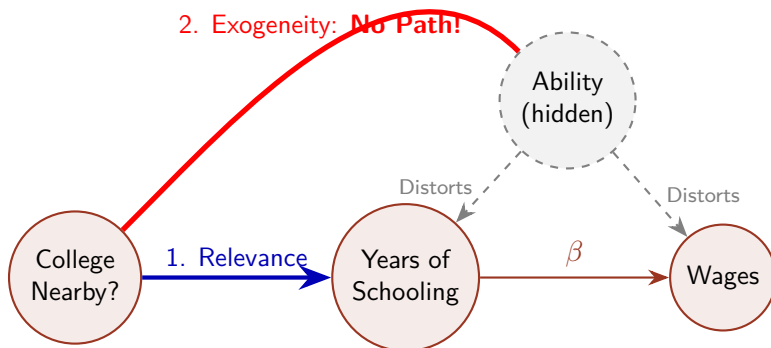
$$\text{Cov}(\text{proximity}, \varepsilon) \approx 0$$

The Intuition

Among equally able people, those who happened to grow up near a college got slightly more education. That **random-ish nudge** lets us isolate the wage effect of schooling alone.

Card (1995): College Proximity as an Instrument

College proximity acts like a “**remote control**” for schooling — it nudges how long people stay in education, but has no direct effect on wages, and no connection to hidden ability.



Why College Proximity Works as an Instrument

- 1 **Relevance:** Living near a college lowers costs of attending $\Rightarrow$ people study longer $\Rightarrow \text{Cov}(\text{proximity}, S) \neq 0$.
- 2 **Exogeneity:** College locations were fixed by **historical decisions** long before today's students were born — unrelated to their ability $\Rightarrow \text{Cov}(\text{proximity}, \epsilon) \approx 0$.

The Research Question

Question: Does serving in the military affect a veteran's earnings later in life?

The ideal experiment: randomly assign some people to the army and leave others at home. Measure wages 10 years later. Done.

The Problem: Selection Bias

In reality, military service is **not random**. Who volunteers or gets drafted depends on:

- Education level (college students could defer)
- Physical health and fitness
- Economic background (poorer men had fewer deferment options)
- Location and political connections

All of these also affect wages. So $\text{Cov}(\text{service}_i, \varepsilon_i) \neq 0$ — OLS gives a **biased** answer.

The Instrument: Vietnam Draft Lottery

Angrist (1990) solved this with a brilliant natural experiment.

In 1969–1972, the US held a **draft lottery**: every birthday of the year was assigned a random number (1–365). Men with low lottery numbers were **called up first**; men with high numbers were largely left alone.

Relevance ✓

Having a **low lottery number** strongly increased the probability of being drafted and serving.

$$\text{Cov}(\underbrace{\text{lottery number}}_Z, \underbrace{\text{served}}_X) \neq 0$$

Easy to verify: just check the first-stage F-stat. It is very large.

Exogeneity ✓

The lottery number was **assigned purely by birthday** — a random draw. It has **no connection** to a person's ability, family background, or anything else that affects wages.

$$\text{Cov}(\text{lottery number}, \varepsilon) = 0$$

This is the whole point: the randomness of the lottery mimics a randomised experiment.

Eligibility as the Instrument: The “Fuzzy” Design

Notice something important: a low lottery number **did not guarantee** service, and a high number **did not guarantee** exemption. Some low-number men got medical deferments; some high-number men volunteered anyway.

This is called a “Fuzzy” IV Design

- The instrument (Z) = **draft eligibility** (lottery number below the cutoff: yes/no)
- The endogenous variable (X) = **actual military service** (did you serve: yes/no)
- Z *shifts the probability* of X but does not determine it perfectly

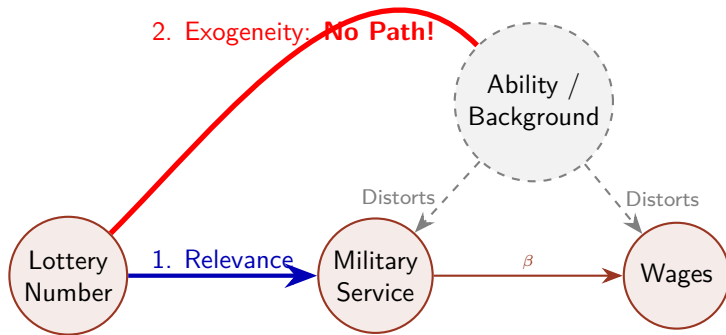
This is exactly the 2SLS setup: Stage 1 regresses actual service on draft eligibility; Stage 2 uses the predicted probability of service to estimate the wage effect.

What IV Estimates Here: The LATE

IV does **not** give the average effect for everyone. It gives the effect for **Compliers** — men who served *because* they got a low lottery number and would not have served otherwise. This is called the **Local Average Treatment Effect (LATE)**.

The Draft Lottery as an Instrument

The lottery number acts like a “remote control” for military service — it nudges who serves, but has no direct path to wages, and no connection to hidden ability or background.



Why the Lottery Works as an Instrument

- ① **Relevance:** Low lottery number $\Rightarrow$ higher probability of being drafted $\Rightarrow \text{Cov}(\text{lottery}, \text{service}) \neq 0$.
- ② **Exogeneity:** The number was assigned **purely by birthday** — it has nothing to do with ability, family wealth, or anything else inside ϵ .

The 2SLS Setup & R Code

Stage 1: Regress actual service on draft eligibility (and controls):

$$\text{served}_i = \pi_0 + \pi_1 \text{eligible}_i + \pi_2 X_i + v_i$$

Stage 2: Regress log wages on the *predicted* probability of service:

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \widehat{\text{served}}_i + \beta_2 X_i + \varepsilon_i$$

```
library(AER)

# draft_eligible = 1 if lottery number below cutoff (the instrument)
# served          = 1 if actually served (endogenous)
# controls: year of birth, race, education

# OLS (biased -- ignores selection)
ols <- lm(log(wage) ~ served + yob + race, data = veterans)

# IV / 2SLS (draft eligibility instruments for actual service)
iv <- ivreg(log(wage) ~ served + yob + race |
            draft_eligible + yob + race,
            data = veterans)

summary(iv, diagnostics = TRUE)
# Check: Weak instruments F >> 10? Wu-Hausman significant?
```

What Did Angrist Find?

Key Results (Angrist 1990)

- **OLS** suggested military service *slightly reduced* wages — but this was likely driven by selection: disadvantaged men (lower earning potential) were more likely to serve.
- **IV estimate** (using draft lottery): serving in Vietnam reduced annual earnings by approximately **15%** for white veterans in the years after service.
- The IV estimate was **larger in magnitude** than OLS — exactly what we expect when negative selection is present (OLS underestimates the harm because it compares already-disadvantaged soldiers to the general population).

Why This Study Became a Classic

It showed that a **historical lottery** — something completely outside anyone's control — could serve as a near-perfect instrument. The draft lottery gave researchers the randomised experiment they could never have run themselves. It launched an entire literature on using **natural experiments** as instruments.

How 2SLS Works: Step by Step

Stage 1: “Clean” the Endogenous Variable

Regress the endogenous X on the instrument Z (and any controls):

$$X_i = \pi_0 + \pi_1 Z_i + v_i$$

Save the **fitted values** $\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i$.

$\hat{X}_i$ contains **only the variation in X driven by Z** — and since Z is exogenous, $\hat{X}_i$ is clean (uncorrelated with ε).

Stage 2: Run the Regression with the Clean Variable

Replace the original X_i with the fitted $\hat{X}_i$:

$$Y_i = \beta_0 + \beta_1 \hat{X}_i + \varepsilon_i^*$$

The estimated $\hat{\beta}_1^{IV}$ is now a **consistent** estimate of the true causal effect.

Warning: Do NOT run 2SLS by hand in two OLS steps!

The second-stage standard errors will be **wrong**. Always use dedicated IV software (e.g., `ivreg` in R).

The IV Estimator: What It Actually Computes

In the simple case with one endogenous variable and one instrument, the IV estimator has a clean closed form:

$$\hat{\beta}_1^{IV} = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)}$$

Intuition: The Wald Estimator

$$\hat{\beta}_1^{IV} = \frac{\text{How much does } Y \text{ move when } Z \text{ moves?}}{\text{How much does } X \text{ move when } Z \text{ moves?}}$$

We use the instrument as a **lever**: we know how hard the lever moves X , and we observe how Y responds. Dividing gives us the $X \rightarrow Y$ effect.

Compare to OLS

OLS uses: $\hat{\beta}_1^{OLS} = \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)}$ — all variation in X , including the dirty part.

When to Use OLS vs IV

	OLS	IV / 2SLS
$\text{Cov}(X, \varepsilon) = 0$	Consistent	Consistent (but wasteful)
$\text{Cov}(X, \varepsilon) \neq 0$	Biased	Consistent
Standard errors	Small	Larger (less precise)
Requires instrument?	No	Yes
Easy to implement?	Yes	Harder

Rule of Thumb

- Start with OLS; ask yourself: *is X exogenous?*
- If yes (randomised experiment, strong theory) $\Rightarrow$ OLS is fine.
- If no (observational data with likely OVB, reverse causality, measurement error) $\Rightarrow$ look for an instrument.

Key Diagnostics for IV

After running 2SLS, you must check two things:

1. First-Stage F-statistic (Test for Weak Instruments)

From Stage 1, check the F-statistic on the excluded instrument(s).

Rule of thumb: $F > 10$ (Stock & Yogo) means the instrument is strong enough. If $F < 10$: your instrument is **weak** — IV estimates become unreliable.

2. Hausman Test (Do We Even Need IV?)

Tests whether OLS and IV give **statistically different answers**.

- H_0 : X is exogenous (OLS is consistent)
- If you reject $H_0 \Rightarrow$ endogeneity confirmed, use IV.
- If you cannot reject $H_0 \Rightarrow$ OLS is probably fine (and more precise).

Exogeneity of Z cannot be tested (usually)

With one instrument and one endogenous variable, we **cannot statistically test** whether $\text{Cov}(Z, \varepsilon) = 0$. With multiple instruments, the **Sargan-Hansen** overidentification test can partially check this.

The Sargan–Hansen Test: Are Your Instruments Valid?

When Can You Use It?

Only when you have **more instruments than endogenous variables** (overidentified model). With one instrument and one endogenous variable, the test **cannot be run** — there is nothing to test against.

What It Tests

H_0 : All instruments are exogenous (uncorrelated with ε).

The idea: if you have two instruments Z_1 and Z_2 , they should both give you the **same answer** for β^{IV} . If they give very different answers, at least one of them is probably correlated with ε — i.e. invalid. The Sargan statistic measures how much the instruments “disagree”.

$$S = n \cdot R_{\varepsilon \sim Z}^2 \sim \chi^2(q - k)$$

where q = number of instruments, k = number of endogenous variables, and R^2 is from regressing the 2SLS residuals on all instruments.

Critical Warning: Rejection vs Non-Rejection

- **Reject H_0** ($p < 0.05$): at least one instrument is **invalid** — your IV results are suspect.
- **Fail to reject H_0** : instruments are **mutually consistent** — but this does *not* prove they are truly exogenous. If *all* your instruments are invalid in the same direction, the test will not catch it. Economic reasoning still required.

Implementing IV in R

```
# Install/load the AER package for IV estimation
# install.packages("AER")
library(AER)

# OLS (for comparison)
ols_model <- lm(log(wage) ~ schooling + experience, data = mydata)
summary(ols_model)

# IV / 2SLS using ivreg()
# Syntax: y ~ endogenous_X + controls | instrument_Z + controls
iv_model <- ivreg(log(wage) ~ schooling + experience |
                  proximity + experience,
                  data = mydata)
summary(iv_model, diagnostics = TRUE)
# diagnostics = TRUE gives:
#   - Weak instruments (First-stage F-stat)
#   - Wu-Hausman test (Is X endogenous?)
#   - Sargan test (Overidentification, if >1 instrument)
```

Key Takeaways

- ➊ **OLS is great** — when $\text{Cov}(X, \varepsilon) = 0$. It is efficient and easy to interpret.
- ➋ **Endogeneity breaks OLS**. It arises from omitted variables, measurement error, or reverse causality, and makes OLS biased even in large samples.
- ➌ **IV / 2SLS is the fix**. A valid instrument Z must:
 - Be **relevant**: correlated with X
 - Be **exogenous**: uncorrelated with ε
- ➍ **2SLS works in two stages**: first “clean” X using Z , then regress Y on $\hat{X}$.
- ➎ **IV is costly**: larger standard errors, harder to find valid instruments, exogeneity is an assumption you must defend with economic arguments.
- ➏ **Always run diagnostics**: first-stage F-stat, Hausman test, Sargan test (if overidentified).

Difference-in-Differences: The Big Idea

Problem: We want the causal effect of a policy/treatment, but treated and untreated groups were **already different** before it happened.

Solution: Compare the *change over time* in the treated group to the *change over time* in the control group.

The DiD Estimator

$$\hat{\delta}^{DiD} = \underbrace{(\bar{Y}_{post}^{treat} - \bar{Y}_{pre}^{treat})}_{\text{change in treated}} - \underbrace{(\bar{Y}_{post}^{control} - \bar{Y}_{pre}^{control})}_{\text{change in control}}$$

The control group tells us what would have happened **without** the treatment (the counterfactual).

Classic Example: Card & Krueger (1994)

New Jersey raised the minimum wage; Pennsylvania did not.

DiD compares employment changes in NJ *relative to* PA — netting out any economy-wide trends.

DiD: Regression Form & the Parallel Trends Assumption

In practice DiD is estimated as a single regression:

$$Y_{it} = \alpha + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \delta (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it}$$

- $\text{Treat}_i = 1$ if unit i is in the treated group
- $\text{Post}_t = 1$ after the policy change
- δ = the DiD estimate (coefficient on the **interaction term**)

The Key Assumption: Parallel Trends

Without treatment, both groups would have followed the **same trend** over time. *This cannot be proven* — you can support it by plotting pre-treatment trends and checking they move together.

How DiD Relates to IV

DiD sidesteps endogeneity by **differencing out** time-invariant confounders. IV instead uses an external instrument. Both are strategies to achieve **as-good-as-random** variation in X — they just find it differently.

GMM: The Idea in Plain English

Recall IV: We exploit the moment condition $\mathbb{E}[Z_i \varepsilon_i] = 0$ (instrument is uncorrelated with the error).

GMM generalises this. Instead of requiring exactly as many instruments as endogenous variables, GMM can handle **many instruments at once** by finding the β that makes the sample moments “as close to zero as possible”.

The GMM Objective

Choose $\hat{\beta}$ to minimise:

$$Q(\beta) = \mathbf{g}(\beta)^\top \mathbf{W} \mathbf{g}(\beta)$$

where $\mathbf{g}(\beta) = \frac{1}{n} \sum_i Z_i \varepsilon_i(\beta)$ are the **sample moment conditions** and $\mathbf{W}$ is a weighting matrix.

Special Cases You Already Know

- $W = (Z^\top Z)^{-1}$, just-identified: GMM $\equiv$ **IV / 2SLS**
- Optimal $W = \hat{S}^{-1}$ (inverse of variance of moments): **Efficient GMM**

IV is just a special case of GMM!

GMM: When to Use It & R Code

Use GMM over 2SLS when...

- You have **more instruments than endogenous variables** (overidentified)
- Errors are **heteroskedastic or autocorrelated** — Efficient GMM adjusts for this automatically, giving smaller standard errors than 2SLS
- You are estimating **non-linear** moment conditions (e.g. dynamic panel models)

```
# install.packages("gmm")
library(gmm)

# Moment condition:  $E[Z * (Y - X*beta)] = 0$ 
# gmm(y ~ x, ~z, data = mydata) where z is your instrument(s)
gmm_model <- gmm(log(wage) ~ schooling + experience,
                  ~ proximity + experience,
                  data = mydata)
summary(gmm_model)
# For 2SLS-equivalent, use: type = "twoStep"
```

Bivariate Probit: When Your Outcome Is Binary

Problem: What if both the endogenous variable X **and** the outcome Y are **binary** (0/1)?

Example: Does having health insurance ($X \in \{0, 1\}$) affect hospitalisation ($Y \in \{0, 1\}$)? Insurance take-up is endogenous (sicker people self-select in).

Using 2SLS with binary outcomes produces **inconsistent** estimates. Instead, use a **Bivariate Probit**.

The Model: Two Probit Equations, One Joint Error

Selection equation (the “first stage”):

$$X_i^* = \gamma_0 + \gamma_1 Z_i + u_i, \quad X_i = \mathbf{1}[X_i^* > 0]$$

Outcome equation (the “second stage”):

$$Y_i^* = \beta_0 + \beta_1 X_i + \eta_i, \quad Y_i = \mathbf{1}[Y_i^* > 0]$$

The key: $(u_i, \eta_i) \sim \text{Bivariate Normal}(0, 0, 1, 1, \rho)$

ρ is the **correlation between the two error terms** — if $\rho \neq 0$, the equations are linked and OLS/single probit is wrong.

Bivariate Probit: How It Relates to IV

IV / 2SLS

Works by **isolating clean variation** in X via instrument Z .

Assumes **linear** relationships.

Breaks down when Y and X are both binary.

Bivariate Probit

Works by **jointly modelling** both equations and estimating ρ directly.

Respects the **binary nature** of both variables.

Still needs an **exclusion restriction** (an instrument Z in the selection equation but not the outcome).

The Common Thread

Both IV and bivariate probit require an **exclusion restriction** — a variable that drives X but has no direct effect on Y . The difference is in **how** they exploit it.

```
# install.packages("sampleSelection")
library(sampleSelection)
bvp <- treatReg(selection = insurance ~ proximity + age,
                outcome   = hospitalised ~ insurance + age,
                data       = mydata)
summary(bvp) # rho = correlation between errors
```

IV Example: Institutions and Growth

Acemoglu, Johnson & Robinson (2001)

Research Question

Do institutions cause economic growth?

Instrument: Settler mortality rates

Causal Chain:

Settler Mortality $\rightarrow$ Colonial Institutions $\rightarrow$ Modern GDP

- High mortality $\Rightarrow$ extractive institutions (no settlement)
- Low mortality $\Rightarrow$ inclusive institutions (permanent settlement)
- Institutions *persist* and shape modern economic development
- Mortality rates do *not* directly affect modern GDP

IV Example: Police and Crime

Levitt (1997)

Research Question

Does hiring more police reduce crime?

Instrument: Electoral cycles

Causal Chain:

Election Timing → Police Hiring → Crime Rates

- Politicians hire more police *before* elections
- Election timing is unrelated to underlying crime trends
- Provides exogenous variation in police force size
- Identifies the causal effect of policing on crime